

CBSE Class 12 Chemistry — Coordination Compounds

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

- Q1.** What is the secondary valency of Cobalt in $[Co(en)_2Cl_2]^+$? (a) 6 (b) 4 (c) 2 (d) 8 [PYQ 2023] [Confidence: Very High]
- Q2.** The counter ion in the coordination compound $[Co(NH_3)_5(NO_2)]Cl_2$ is: (a) Ammine (b) Cobalt (c) Chloride (d) Nitro [PYQ SQP 2025-26] [Confidence: Very High]
- Q3.** The formula of the complex dichloridobis(ethane-1,2-diamine)platinum(IV) nitrate is: (a) $[PtCl_2(en)_2(NO_3)_2]$ (b) $[PtCl_2(en)_2](NO_3)_2$ (c) $[PtCl_2(en)_2(NO_3)]NO_3$ (d) $[Pt(en)_2(NO_3)_2]Cl_2$ [PYQ 2023] [Confidence: Very High]
- Q4.** Predict the number of ions produced per formula unit in an aqueous solution of $[Co(en)_3]Cl_3$: (a) 4 (b) 3 (c) 6 (d) 2 [PYQ 2019] [Confidence: High]
- Q5.** The oxidation number of Ni in $[Ni(CO)_4]$ is: (a) 0 (b) 2 (c) 3 (d) 4 [PYQ 2020] [Confidence: Very High]
- Q6.** The coordination number of Co in the complex $[Co(en)_3]^{3+}$ is: (a) 3 (b) 6 (c) 4 (d) 5 [PYQ 2020] [Confidence: Very High]
- Q7.** The CFSE for octahedral $[CoCl_6]^{4-}$ is $18,000\text{ cm}^{-1}$. The CFSE for tetrahedral $[CoCl_4]^{2-}$ will be: (a) $18,000\text{ cm}^{-1}$ (b) $16,000\text{ cm}^{-1}$ (c) $8,000\text{ cm}^{-1}$ (d) $20,000\text{ cm}^{-1}$ [PYQ SQP 2022] [Confidence: High]
- Q8.** The crystal field splitting energy for octahedral (Δ_o) and tetrahedral (Δ_t) complexes are related as: (a) $\Delta_t = \frac{4}{9}\Delta_o$ (b) $\Delta_t = \frac{1}{2}\Delta_o$ (c) $\Delta_o = 2\Delta_t$ (d) $\Delta_o = \frac{4}{9}\Delta_t$ [PYQ 2020] [Confidence: Very High]
- Q9.** Ambidentate ligands like NO_2^- and SCN^- are: (a) unidentate (b) didentate (c) polydentate (d) has variable denticity [PYQ SQP 2019] [Confidence: High]
- Q10.** Is the central metal atom in coordination complexes a Lewis acid or a Lewis base? (a) Lewis Acid (b) Lewis Base (c) Both (d) Neither [PYQ SQP 2023] [Confidence: Very High]
- Q11.** The number of ions formed on dissolving one molecule of $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ (Mohr's salt) in water is: (a) 3 (b) 4 (c) 5 (d) 6 [PYQ SQP 2022] [Confidence: Medium]
- Q12.** Assertion (A): Linkage isomerism arises in coordination compounds because of ambidentate ligands. Reason (R): Ambidentate ligand like NO_2 has two different donor atoms i.e., N and O. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2020] [Confidence: Very High]

Q13. Assertion (A): $[Pt(en)_2Cl_2]^{2+}$ complex is less stable than $[Pt(NH_3)_4Cl_2]^{2+}$ complex. Reason (R): $[Pt(en)_2Cl_2]^{2+}$ complex shows chelate effect. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2020] [Confidence: High]

Q14. What type of isomerism is exhibited by the complex $[Co(NH_3)_5(SCN)]^{2+}$? [PYQ 2017] [Confidence: Very High]

Q15. In a coordination entity, the electronic configuration of the central metal ion is $t_{2g}^3 e_g^1$. Is the coordination compound a high spin or low spin complex? [PYQ SQP Term-2 2022] [Confidence: High]

Q16. Does ionization isomer for the following compound exist? $Hg[Co(SCN)_4]$ [PYQ SQP 2023] [Confidence: High]

Q17. Why are low spin tetrahedral complexes rarely observed? [PYQ 2017] [Confidence: Medium]

Q18. One mole of $CrCl_3 \cdot 6H_2O$ compound reacts with excess $AgNO_3$ solution to yield two moles of $AgCl(s)$. What is the structural formula of the compound? [PYQ 2020] [Confidence: Very High]

Q19. Write the coordination number and oxidation state of Platinum in the complex $[Pt(en)_2Cl_2]^{2+}$. [PYQ 2018 / Practice] [Confidence: Very High]

Q20. What is the denticity of the ligand EDTA? [CBSE PDF] [Confidence: High]

2-Mark Questions (Short Answer)

Q21. What is a chelate complex? Give one example. [PYQ 2023] [Confidence: Very High]

Q22. What are heteroleptic complexes? Give one example. [PYQ 2023] [Confidence: Very High]

Q23. Explain ambidentate ligand with the help of a suitable example. [Delhi Gov. SQP Term-2 2022] [Confidence: Very High]

Q24. What is the difference between a coordination complex and a double salt? Give an example of each. [PYQ 2019] [Confidence: High]

Q25. Give the formulae of the following compounds: (A) Potassium tetrahydroxidozincate (II) (B) Hexaammineplatinum (IV) chloride [PYQ 2020] [Confidence: High]

Q26. Write the IUPAC names of the following: (A) $[Co(NH_3)_5(ONO)]^{2+}$ (B) $K_2[NiCl_4]$ [PYQ 2023] [Confidence: Very High]

Q27. What type of isomerism is shown by the complex $[Co(NH_3)_6][Cr(CN)_6]$? Write the structure of its possible isomer. [PYQ 2017] [Confidence: High]

Q28. Explain why $[NiCl_4]^{2-}$ is paramagnetic but $[Ni(CN)_4]^{2-}$ is diamagnetic. [PYQ 2017] [Confidence: Very High]

Q29. Draw the geometrical isomers of the complex $[Pt(en)_2Cl_2]^{2+}$. Which of these isomers is optically active? [PYQ 2016] [Confidence: Very High]

Q30. What is meant by the "spectrochemical series"? What is the basic difference between a weak field ligand and a strong field ligand? [CBSE Term-2 2022] [Confidence: High]

Q31. Using crystal field theory, write the electronic configuration of a d^5 ion in an octahedral field when: (i) $\Delta_o > P$ (ii) $\Delta_o < P$ [CBSE Term-2 2022] [Confidence: Very High]

Q32. Explain the concept of synergic bonding in metal carbonyls with a brief description. [Practice] [Confidence: Medium]

3-Mark Questions (Short Answer)

Q33. Using Valence Bond Theory, explain the following in relation to the paramagnetic complex $[Mn(CN)_6]^{3-}$: (A) Type of hybridisation (B) Magnetic moment value (C) Type of complex (inner or outer orbital). [PYQ SQP 2022] [Confidence: Very High]

Q34. Two complexes are given as $[CoF_6]^{3-}$ and $[Co(en)_3]^{3+}$. Identify which complex is: (A) diamagnetic (B) outer orbital complex (C) low spin complex (Atomic no. of Co = 27) [PYQ 2019] [Confidence: High]

Q35. Explain why $[Ni(H_2O)_6]^{2+}$ (aq) is green in colour whereas $[Ni(H_2O)_4(en)]^{2+}$ (aq) is blue in colour. Give reason in support of your answer. [CBSE SQP Term-2 2022] [Confidence: High]

Q36. The formula $Co(NH_3)_5CO_3Cl$ could represent a carbonate or a chloride. Write the structures and IUPAC names of the two possible isomers. [PYQ SQP 2022] [Confidence: Very High]

Q37. Write the state of hybridisation, shape, and IUPAC name of the following complex: $[Ni(CN)_4]^{2-}$. [PYQ 2014] [Confidence: Very High]

Q38. Using IUPAC norms write the formulae for the following: (A) Sodium dicyanidoaurate (I) (B) Tetraamminechloridonitrito-N-platinum (IV) sulphate Write the type of structural isomerism exhibited by complex (B). [PYQ 2017] [Confidence: High]

Q39. Write the IUPAC name, stereochemistry, and magnetic behaviour of the complex $K_4[Mn(CN)_6]$. (At. no. Mn = 25). [PYQ 2014] [Confidence: Medium]

Q40. Write the IUPAC name of the complex $[Cr(NH_3)_4Cl_2]^+$. Draw the structures of the geometrical isomers for this complex. [PYQ 2019] [Confidence: Very High]

Q41. Give the formula of the following coordination entity: A Co^{3+} ion is bound to one Cl^- , one NH_3 molecule and two bidentate ethylene diamine (en) molecules. Write the IUPAC name and magnetic behaviour of this coordination entity. [PYQ 2019] [Confidence: High]

Q42. Why is a solution of $[Ni(H_2O)_6]^{2+}$ green while a solution of $[Ni(CN)_4]^{2-}$ is colourless? [PYQ 2017] [Confidence: High]

5-Mark Questions (Long Answer / Case-Study)

Q43. Case Study: Crystal Field Splitting by Various Ligands Transition metal complexes show distinct colours depending on the nature of ligands attached to the central metal ion. The crystal field splitting energy (Δ_o) dictates whether a complex will absorb light in the visible region. Consider four ligands X, Y, Z, W attached to Cr^{3+} forming octahedral complexes $[\text{CrA}_6]^{3\pm}$. The absorbed wavelengths are: X = 750 nm, Y = 600 nm, Z = 450 nm, W = 400 nm. (A) Out of the ligands X, Y, Z, and W, which ligand causes maximum crystal field splitting (Δ_o)? Why? (1.5 Marks) (B) Which of these ligands is the weakest field ligand? Why? (1.5 Marks) (C) If the ligands attached to Cr^{3+} are H_2O , CN^- , Cl^- , and NH_3 (not in this order), assign the correct ligand to Z and W. (2 Marks) [PYQ SQP 2023 / Practice] [Confidence: Very High]

Q44. (A) Write the state of hybridisation, shape, IUPAC name, and magnetic behavior of the complex $[\text{Co}(\text{NH}_3)_6]^{3+}$. (Atomic no. of Co = 27). (3 Marks) (B) Explain, using Valence Bond Theory (VBT), why $[\text{NiCl}_4]^{2-}$ has a tetrahedral geometry while $[\text{Ni}(\text{CN})_4]^{2-}$ has a square planar geometry. (2 Marks) [PYQ 2014 / Practice] [Confidence: Very High]

Q45. (A) Define geometrical and optical isomerism in coordination compounds. (2 Marks) (B) Draw the structures of the enantiomers (optical isomers) of $[\text{Co}(\text{en})_3]^{3+}$. (2 Marks) (C) Draw the *cis* and *trans* forms of the complex $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$. (1 Mark) [Practice] [Confidence: High]

Q46. (A) State the main postulates of Werner's theory of coordination compounds. (2 Marks) (B) Draw the crystal field splitting diagram for d-orbitals in an octahedral crystal field. Clearly label the t_{2g} and e_g energy levels and the splitting energy Δ_o . (2 Marks) (C) How does the magnitude of Δ_o affect the electronic configuration of a d^4 metal ion if it is a strong field ligand? (1 Mark) [PYQ 2017 / Practice] [Confidence: Very High]

Q47. (A) Using IUPAC norms, write the formulas for the following: (i) Pentaamminenitrito-O-cobalt(III) chloride (ii) Potassium tetracyanonickelate(II) (iii) Tris(ethane-1,2-diamine)chromium(III) chloride (3 Marks) (B) Identify the oxidation state of the central metal ion in each of the above complexes. (1 Mark) (C) Predict the coordination number of the central metal ion in complex (iii). (1 Mark) [PYQ 2019 / 2017] [Confidence: High]

Q48. (A) When a coordination compound $\text{CoCl}_3 \cdot 6\text{NH}_3$ is mixed with AgNO_3 , 3 moles of AgCl are precipitated per mole of the compound. Write the structural formula of the complex. (1 Mark) (B) What is the secondary valency of Cobalt in this complex? (1 Mark) (C) Using VBT, discuss the bonding, geometry, and magnetic property of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$. (Given atomic number of Fe = 26). (3 Marks) [PYQ 2016 / Practice] [Confidence: Very High]

Q49. (A) Using VBT, discuss the bonding, geometry, and magnetic property of $[\text{Fe}(\text{CN})_6]^{3-}$. (Given atomic number of Fe = 26). (3 Marks) (B) Write the IUPAC names for: (i) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (ii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ (2 Marks) [PYQ 2016 / Practice] [Confidence: High]

Q50. (A) Define the following terms with one example each: (i) Coordination polyhedron (ii) Homoleptic complex (2 Marks) (B) Give the importance/application of coordination compounds in: (i) Biological systems (give one example) (ii) Medicine (give one example) (2

Marks) (C) What is the denticity of the $C_2O_4^{2-}$ (oxalate) ligand? (1 Mark) [Practice based on CBSE syllabus] [Confidence: High]

Answer Key

Q1. (a) 6. (ethylenediamine is bidentate: $2 \times 2 = 4$, plus two chlorides = 6). **Q2.** (c) Chloride. (The ions outside the square bracket). **Q3.** (b) $[PtCl_2(en)_2](NO_3)_2$. (Platinum IV means 2 Cl^- and 2 NO_3^- balance the +4 to zero overall charge). **Q4.** (a) 4. (One $[Co(en)_3]^{3+}$ ion and three Cl^- ions). **Q5.** (a) 0. (CO is a neutral ligand). **Q6.** (b) 6. (en is a bidentate ligand, $3 \times 2 = 6$). **Q7.** (c) $8,000\text{ cm}^{-1}$. ($\Delta_t = \frac{4}{9} \times 18000 = 8000$). **Q8.** (a) $\Delta_t = \frac{4}{9} \Delta_o$. **Q9.** (a) unidentate. (They bind through only one donor atom at a time). **Q10.** (a) Lewis Acid. (It accepts electron pairs from ligands). **Q11.** (c) 5. (Dissociates into 1 Fe^{2+} , 2 NH_4^+ , and 2 SO_4^{2-}). **Q12.** (a) Both A and R are true and R is the correct explanation of A. **Q13.** (d) A is false but R is true. ($[Pt(en)_2Cl_2]^{2+}$ is MORE stable due to the chelate effect). **Q14.** Linkage isomerism. (Due to ambidentate SCN ligand). **Q15.** High spin. ($t_{2g}^3 e_g^1$ indicates electrons are placed in e_g before pairing in t_{2g} , meaning $\Delta_o < P$). **Q16.** No. There are no exchangeable counter ions outside the coordination sphere to swap with ligands. **Q17.** Because the crystal field splitting energy for tetrahedral complexes (Δ_t) is naturally smaller than the pairing energy (P). **Q18.** $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$. (Since 2 moles of AgCl precipitate, 2 Cl^- must be outside the coordination sphere). **Q19.** Coordination number = 6 (en is bidentate). Oxidation state = +4 ($x + 0 + 2(-1) = +2 \rightarrow x = +4$). **Q20.** Hexadentate. (It can bind through two nitrogen and four oxygen atoms).

Q21. A complex in which a multidentate ligand bonds to the central metal atom forming a ring structure. Example: $[Co(en)_3]^{3+}$. **Q22.** Complexes in which a metal is bound to more than one kind of donor groups/ligands. Example: $[Co(NH_3)_4Cl_2]^+$. **Q23.** Ligands that can ligate through two different atoms present in them. Example: NO_2^- can bind through N (nitrito-N) or O (nitrito-O). **Q24.** Double salts completely dissociate into simple ions in water (e.g., Mohr's salt). Coordination complexes do not completely dissociate and retain their identity in solution (e.g., $K_4[Fe(CN)_6]$). **Q25.** (A) $K_2[Zn(OH)_4]$ (B) $[Pt(NH_3)_6]Cl_4$. **Q26.** (A) Pentaamminenitrito-O-cobalt(III) ion. (B) Potassium tetrachloridonickelate(II). **Q27.** Coordination isomerism. Isomer: $[Cr(NH_3)_6][Co(CN)_6]$. **Q28.** In $[NiCl_4]^{2-}$, Cl^- is a weak field ligand; no pairing of $3d^8$ electrons occurs, leaving 2 unpaired electrons (paramagnetic). In $[Ni(CN)_4]^{2-}$, CN^- is a strong field ligand, forcing electron pairing, leaving 0 unpaired electrons (diamagnetic). **Q29.** cis and trans isomers. The cis isomer is optically active because it lacks a plane of symmetry. (Trans has a plane of symmetry). **Q30.** A series in which ligands are arranged in increasing order of their crystal field splitting magnitude. Weak field ligands produce small splitting ($\Delta_o < P$), strong field produce large splitting ($\Delta_o > P$). **Q31.** (i) If $\Delta_o > P$ (strong field): $t_{2g}^5 e_g^0$. (ii) If $\Delta_o < P$ (weak field): $t_{2g}^3 e_g^2$. **Q32.** In metal carbonyls, a σ -bond forms by donation of lone pair from CO carbon to the empty metal d-orbital. A π -backbond forms by donation of electrons from the filled metal d-orbital into the empty antibonding π^* orbital of CO. This mutually strengthens the metal-ligand bond.

Q33. (A) d^2sp^3 (strong field CN^-). (B) Mn^{3+} is $3d^4$, forced pairing gives $t_{2g}^4e_g^0$, 2 unpaired electrons. $\mu = \sqrt{2(4)} = \sqrt{8} = 2.83$ BM. (C) Inner orbital complex. **Q34.** (A) Diamagnetic: $[Co(en)_3]^{3+}$ (en is strong field, d^6 pairs up). (B) Outer orbital: $[CoF_6]^{3-}$ (uses $4d, sp^3d^2$). (C) Low spin: $[Co(en)_3]^{3+}$. **Q35.** 'en' is a stronger field ligand than H_2O . It increases Δ_o , thereby the complex absorbs higher energy (shorter wavelength) light for d-d transition, transmitting a different complementary color (blue instead of green). **Q36.** (1) $[Co(NH_3)_5CO_3]Cl$: Pentaamminecarbonatocobalt(III) chloride. (2) $[Co(NH_3)_5Cl]CO_3$: Pentaamminechloridocobalt(III) carbonate. **Q37.** Hybridisation: dsp^2 . Shape: Square planar. Name: Tetracyanonickelate(II) ion. **Q38.** (A) $Na[Au(CN)_2]$. (B) $[Pt(NH_3)_4Cl(NO_2)]SO_4$. Complex (B) exhibits linkage isomerism (due to NO_2) and ionization isomerism (with SO_4). **Q39.** Name: Potassium hexacyanidomanganate(II). Stereochem: Octahedral. Magnetic: Low spin, paramagnetic (1 unpaired electron, d^5 strong field). **Q40.** Tetraamminedichloridochromium(III) ion. Isomers: Draw cis (Cl adjacent) and trans (Cl opposite) octahedral structures. **Q41.** Formula: $[Co(en)_2(NH_3)Cl]^{2+}$. Name: Amminechloridobis(ethane-1,2-diamine)cobalt(III) ion. Magnetic: Diamagnetic (Co^{3+} with strong en/NH_3 pairs d^6). **Q42.** H_2O is a weak field ligand, leaving unpaired electrons in Ni^{2+} (d^8) which undergo d-d transitions, emitting green light. CN^- is strong field, pairing electrons in dsp^2 hybridization; no unpaired electrons means no d-d transition (colourless). **Q43.** (A) Ligand W. Highest splitting energy means it absorbs the highest energy light (shortest wavelength, 400 nm). (B) Ligand X. Smallest splitting absorbs lowest energy light (longest wavelength, 750 nm). (C) Z (450 nm) = NH_3 (stronger than Cl^- , weaker than CN^-). W (400 nm) = CN^- (strongest field). **Q44.** (A) d^2sp^3 , Octahedral, Hexaamminecobalt(III) ion, Diamagnetic. (B) In $[NiCl_4]^{2-}$, Cl^- is weak field, no pairing, uses sp^3 (tetrahedral). In $[Ni(CN)_4]^{2-}$, CN^- is strong field, pairs $3d^8$ leaving one empty 3d orbital, uses dsp^2 (square planar). **Q45.** (A) Geometrical: Isomers having different spatial arrangements of ligands (e.g., cis/trans). Optical: Non-superimposable mirror images (enantiomers) that rotate plane-polarized light. (B) Draw mirror images of $[Co(en)_3]^{3+}$ propeller structures. (C) Draw octahedral complex with Cl adjacent (cis) and Cl opposite 180 degrees (trans). **Q46.** (A) Metals possess primary valency (oxidation state, ionizable) and secondary valency (coordination number, non-ionizable, directional). (B) Draw 5 degenerate d-orbitals splitting into 3 lower (t_{2g}) and 2 upper (e_g). Label gap as Δ_o . (C) Strong field ligand: $\Delta_o > P$, so the 4th electron pairs in t_{2g} rather than entering e_g . Configuration: $t_{2g}^4e_g^0$. **Q47.** (A) (i) $[Co(NH_3)_5(ONO)]Cl_2$ (ii) $K_2[Ni(CN)_4]$ (iii) $[Cr(en)_3]Cl_3$. (B) (i) +3, (ii) +2, (iii) +3. (C) 6. **Q48.** (A) $[Co(NH_3)_6]Cl_3$. (B) Secondary valency = 6. (C) Fe^{3+} is $3d^5$. H_2O is weak field, no pairing. Hybridization is sp^3d^2 . Geometry is outer-orbital octahedral. Highly paramagnetic (5 unpaired electrons). **Q49.** (A) Fe^{3+} is $3d^5$. CN^- is strong field, forces pairing. Leaves 1 unpaired electron. Uses inner 3d orbitals (d^2sp^3). Geometry: Octahedral. Paramagnetic. (B) (i) Potassium trioxalatoferrate(III). (ii) Diamminechloridonitrito-N-platinum(II). **Q50.** (A) (i) Polyhedron: Spatial arrangement of ligands defining geometry (e.g., octahedral). (ii) Homoleptic: Complex with only one kind of donor group (e.g., $[Ni(CO)_4]$). (B) (i) Chlorophyll is a Mg complex, Hemoglobin is an Fe complex. (ii) Cisplatin used in cancer treatment. (C) Bidentate (denticity = 2).

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).